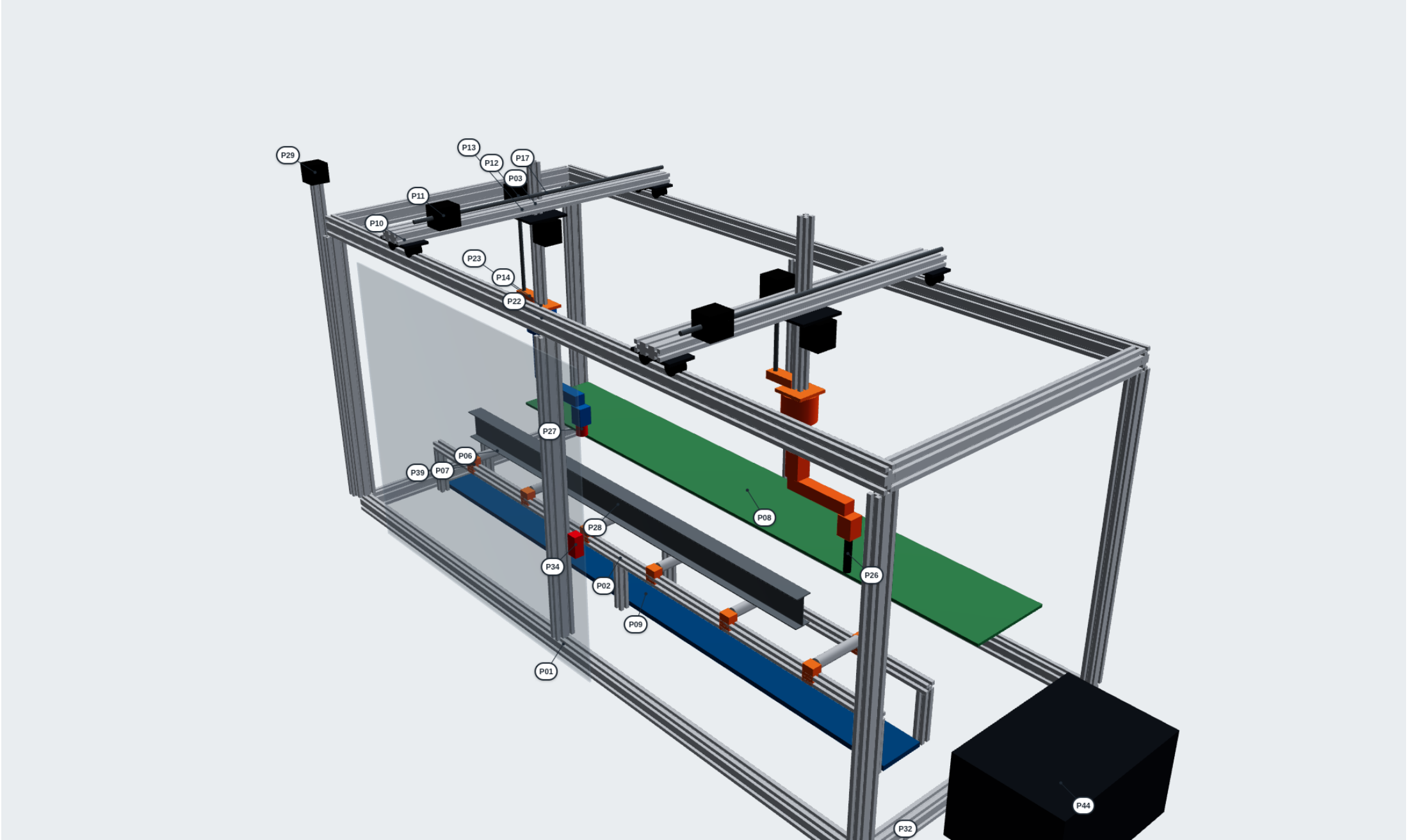


General assembly

Each balloon is a part number (P01 ...) - the same number as the shopping list and the CAD. The table lists every part in the model and how many there are.



Part	Name	No.
P01	V-slot 20x40 aluminium extrusion	14
P02	V-slot 20x20 aluminium extrusion	11
P03	V-slot 20x60 aluminium extrusion	2
P06	Bed rollers	6
P07	Roller brackets	12
P08	Outfeed table board	1
P09	Scrap tray	1
P10	V-wheel gantry plate kits	14
P11	NEMA 17 stepper motor	4
P12	NEMA 17 with integrated T8 lead screw	4
P13	Z motor bracket (3D printed)	2
P14	Lead-screw nut block (3D printed)	2
P17	8 mm cross shafts + pillow blocks	2
P22	SO-101 6-servo arm kit (follower)	8
P23	Arm mount plate (3D printed)	2
P26	Cutter 'torch' for the prototype	1
P27	Handler magnet	1
P28	Model beams	1
P29	USB wide-angle camera (safety zone)	1
P32	Emergency stop station	2
P34	Door interlock switch	1
P39	Polycarbonate sheet 3 mm	1
P44	Control box	1

Steel Beam Cutting Cell - 1:5 prototype
Sheet: general assembly

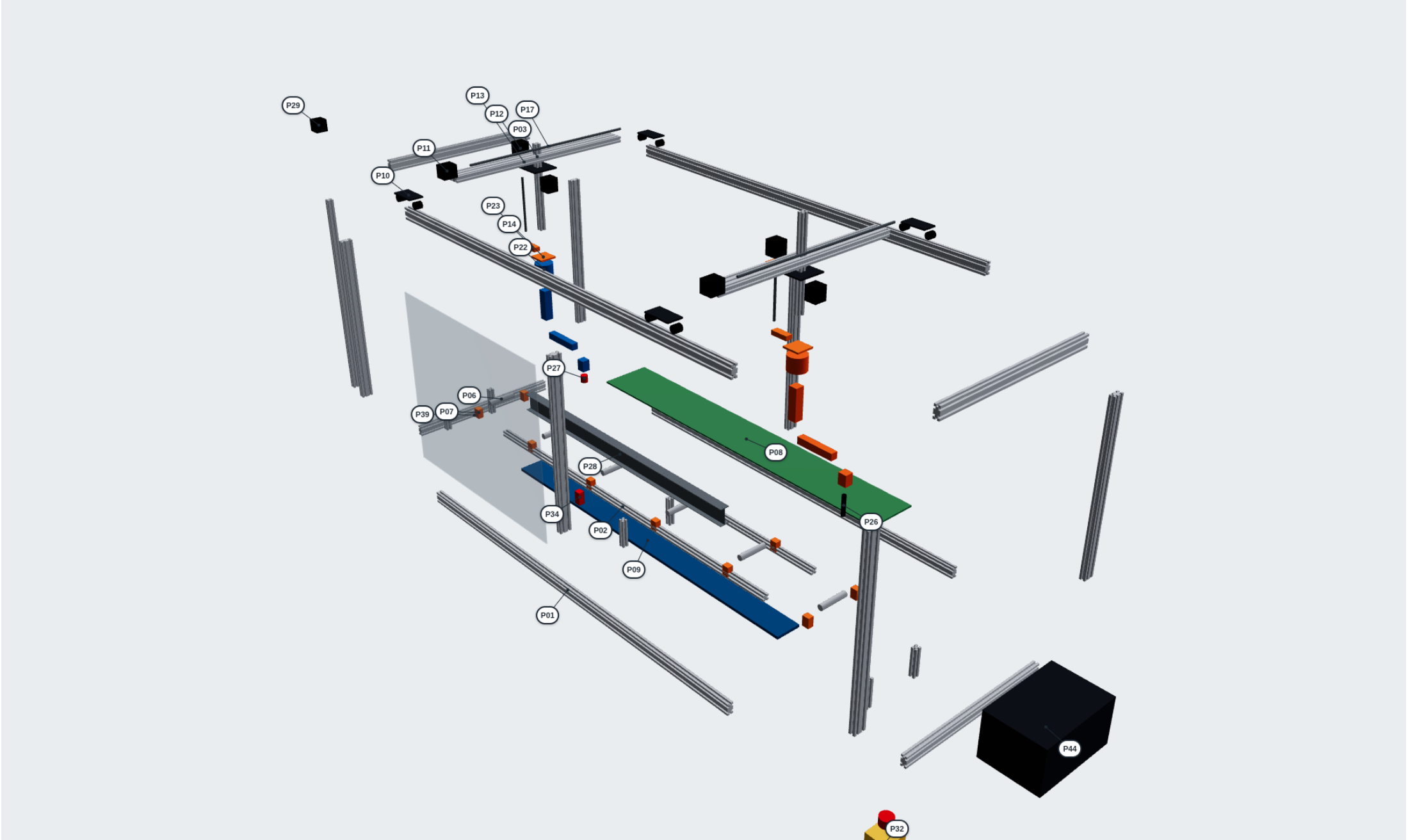
Drawing: BC-P1-GA

Drawn: Ravi Mahadeva
Date: 04 Oct 2026 Units: mm Scale: not to scale

Sheet 1
CAD: cad/prototype_1to5.step

Exploded view - what goes where

The parts pulled apart from the middle so you can see how they fit together.



Part	Name	No.
P01	V-slot 20x40 aluminium extrusion	14
P02	V-slot 20x20 aluminium extrusion	11
P03	V-slot 20x60 aluminium extrusion	2
P06	Bed rollers	6
P07	Roller brackets	12
P08	Outfeed table board	1
P09	Scrap tray	1
P10	V-wheel gantry plate kits	14
P11	NEMA 17 stepper motor	4
P12	NEMA 17 with integrated T8 lead screw	4
P13	Z motor bracket (3D printed)	2
P14	Lead-screw nut block (3D printed)	2
P17	8 mm cross shafts + pillow blocks	2
P22	SO-101 6-servo arm kit (follower)	8
P23	Arm mount plate (3D printed)	2
P26	Cutter 'torch' for the prototype	1
P27	Handler magnet	1
P28	Model beams	1
P29	USB wide-angle camera (safety zone)	1
P32	Emergency stop station	2
P34	Door interlock switch	1
P39	Polycarbonate sheet 3 mm	1
P44	Control box	1

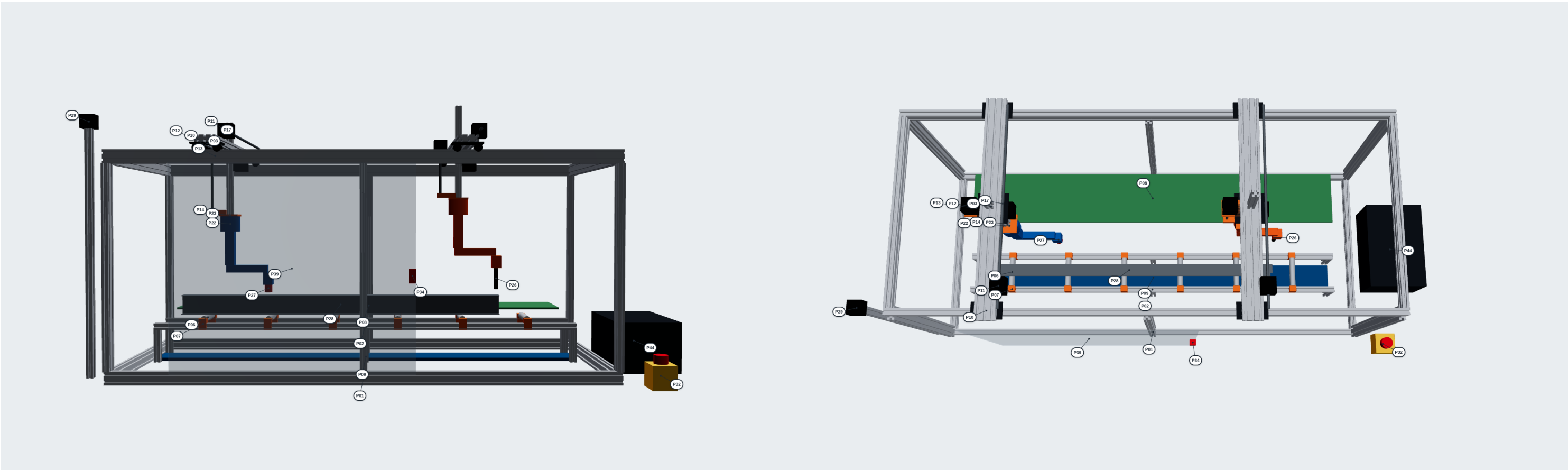
Steel Beam Cutting Cell - 1:5 prototype
Sheet: exploded view

Drawing: BC-P1-GA

Drawn: Ravi Mahadeva
Date: 04 Oct 2026 Units: mm Scale: not to scale

Sheet 2
CAD: cad/prototype_1to5.step

Front view and top view



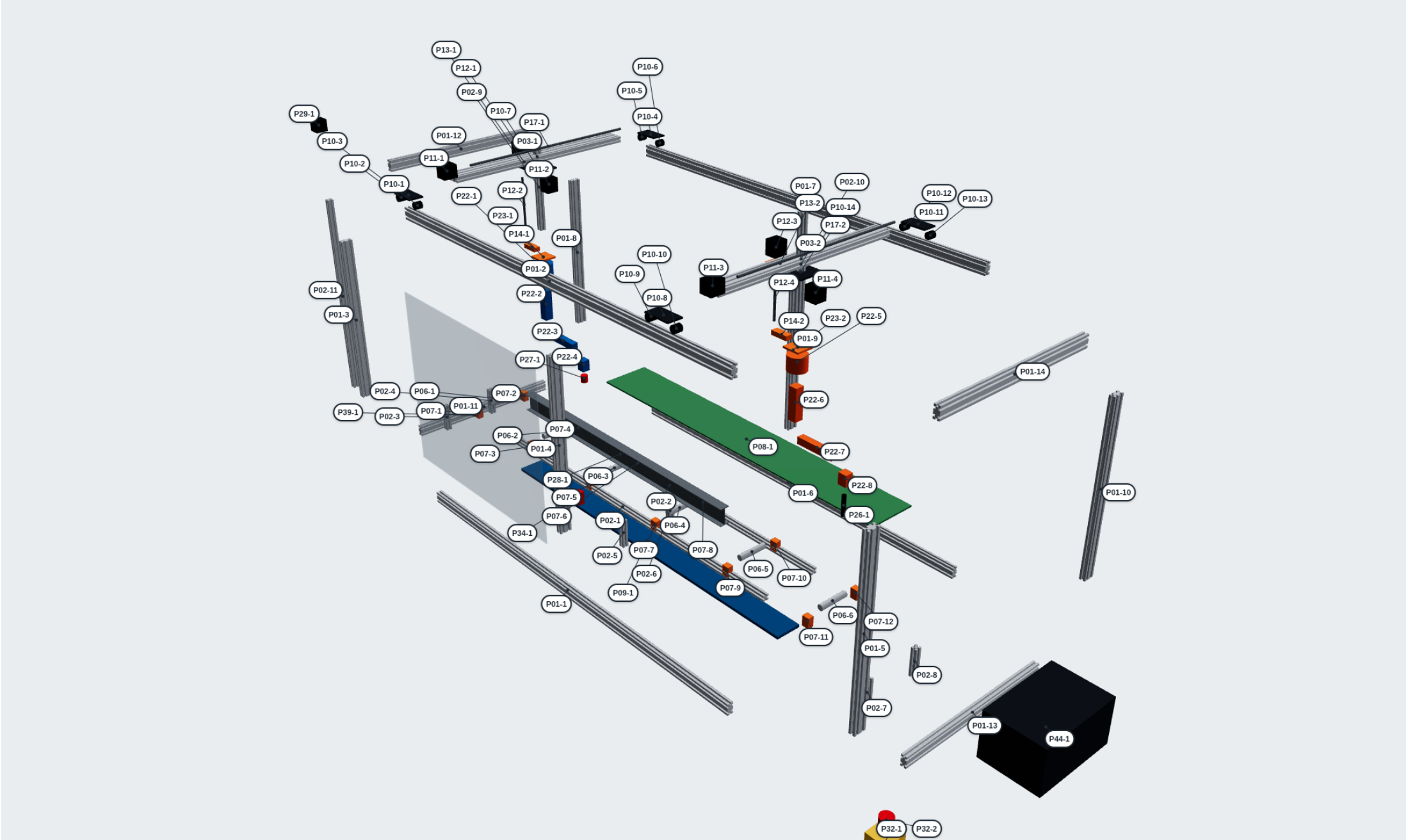
Front - from the operator's side

Top - looking down

Steel Beam Cutting Cell - 1:5 prototype Sheet: front and top views	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 3 CAD: cad/prototype_1to5.step

Every position

Each balloon is a position: part number - copy (P11-2 = the second NEMA 17 motor). The list on the following sheets says what each position is and where it goes.



Steel Beam Cutting Cell - 1:5 prototype Sheet: every position	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 4 CAD: cad/prototype_1to5.step

Parts list - part numbers, positions and where to buy

Part	Name	What to look for	Positions in the model	Buy	Each	Where to buy
P01	V-slot 20x40 aluminium extrusion	cut to size: 4 x 1500, 6 x 600, 4 x 580 mm (cad/prototype_cut_list.csv)	P01-1, P01-2, P01-3, P01-4, P01-5, P01-6, P01-7, P01-8, P01-9, P01-10, P01-11, P01-12, P01-13, P01-14	11.9 m	£14	Ooznest (cut to size), Amazon UK product page
P02	V-slot 20x20 aluminium extrusion	cut to size: 2 x 1300, 2 x 300, 6 x 100, 1 x 720 mm	P02-1, P02-2, P02-3, P02-4, P02-5, P02-6, P02-7, P02-8, P02-9, P02-10, P02-11	4.5 m	£8	Ooznest search shop
P03	V-slot 20x60 aluminium extrusion	cut to size: 2 x 660 mm	P03-1, P03-2	2	£16	Ooznest search shop
P04	Corner brackets + M5 T-nuts + M5 bolts	40 cast corner brackets, 200 drop-in T-nuts, M5x8/M5x10 button heads	<i>not drawn (wiring, inside the control box, tools)</i>	1 kit	£45	Ooznest, Amazon UK search shop
P05	Rubber feet / levelling feet	M8 levelling feet for the base rails	<i>not drawn (wiring, inside the control box, tools)</i>	4	£3	Amazon UK search shop
P06	Bed rollers	20 mm aluminium tube x 100 mm on 8 mm silver steel shaft, 2 x 608 bearings each	P06-1, P06-2, P06-3, P06-4, P06-5, P06-6	6	£4	Amazon UK, Simply Bearings search shop
P07	Roller brackets	3D printed in PETG (12 off)	P07-1, P07-2, P07-3, P07-4, P07-5, P07-6, P07-7, P07-8, P07-9, P07-10, P07-11, P07-12	12	£0.50	print <i>make / print it</i>
P08	Outfeed table board	6 mm plywood or MDF 1300 x 180 mm	P08-1	1	£8	builders' merchant search shop
P09	Scrap tray	folded aluminium sheet or 3D printed tray 1300 x 90 mm	P09-1	1	£10	print / sheet <i>make / print it</i>
P10	V-wheel gantry plate kits	20-series gantry plate + 4 V-wheels (625 bearings) + eccentric spacers	P10-1, P10-2, P10-3, P10-4, P10-5, P10-6, P10-7, P10-8, P10-9, P10-10, P10-11, P10-12, P10-13, P10-14	6	£18	Ooznest search shop
P11	NEMA 17 stepper motor	1.5-2 A, 45-48 mm body, 5 mm shaft	P11-1, P11-2, P11-3, P11-4	4	£12	Ooznest, StepperOnline search shop
P12	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead, 300 mm screw, with brass anti-backlash nut	P12-1, P12-2, P12-3, P12-4	2	£25	StepperOnline, Amazon UK search shop
P13	Z motor bracket (3D printed)	PETG, holds the Z motor on the carriage plate	P13-1, P13-2	2	£0	print <i>make / print it</i>
P14	Lead-screw nut block (3D printed)	PETG block with the brass T8 nut (comes with the motor), fixed to the Z axis	P14-1, P14-2	2	£0	print <i>make / print it</i>
P15	GT2 belt 6 mm	10 m reel (8 m needed: 2 per bridge along the rails, 1 per carriage)	<i>not drawn (wiring, inside the control box, tools)</i>	1	£12	Ooznest search shop
P16	GT2 20T pulleys + idlers	4 x 20T 8 mm bore (cross shafts), 2 x 20T 5 mm bore (Y), 8 idlers	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£25	Ooznest search shop
P17	8 mm cross shafts + pillow blocks	2 x 700 mm 8 mm silver steel, 4 x KFL08 bearings, 2 x 5-8 mm couplers	P17-1, P17-2	1 set	£25	Amazon UK, Simply Bearings search shop
P18	Home / limit switches	mechanical microswitch with lever, NC wiring	<i>not drawn (wiring, inside the control box, tools)</i>	8	£1.50	Amazon UK search shop
P19	FluidNC 6-axis CNC controller (ESP32)	6 driver sockets, USB to the Jetson (e.g. Elecrow / Bart Dring 6x CNC Controller)	<i>not drawn (wiring, inside the control box, tools)</i>	1	£40	Elecrow, Tindie product page
P20	TMC2209 stepper driver modules	StepStick format, UART or standalone	<i>not drawn (wiring, inside the control box, tools)</i>	6	£5	Amazon UK, Ooznest search shop
P21	Shielded 4-core motor cable + drag chain	10 m 4 x 0.5 mm2 shielded, 2 m 10x15 drag chain	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£30	Amazon UK, RS search shop
P22	SO-101 6-servo arm kit (follower)	6 x Feetech STS3215 serial-bus servos + bus servo driver board; print the arm parts in PLA	P22-1, P22-2, P22-3, P22-4, P22-5, P22-6, P22-7, P22-8	2	£240	Unmanned Tech UK (servo kit Pro, unprinted), Seed product page
P23	Arm mount plate (3D printed)	PETG plate: bolts the arm base under the Z axis	P23-1, P23-2	2	£0	print <i>make / print it</i>
P24	Bus servo adapter	Waveshare Bus Servo Adapter (A) - USB to the Jetson, one per arm	<i>not drawn (wiring, inside the control box, tools)</i>	2	£5	Waveshare, The Pi Hut product page
P25	12 V power supply for the servos	Mean Well LRS-100-12 (12 V 8.5 A) - check your servos are the 12 V version	<i>not drawn (wiring, inside the control box, tools)</i>	1	£16	RS, Farnell search shop
P26	Cutter 'torch' for the prototype	pen in a sprung 3D-printed holder (marks every cut line on the model beam)	P26-1	1	£3	print <i>make / print it</i>

Steel Beam Cutting Cell - 1:5 prototype Sheet: parts list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 5 CAD: cad/prototype_1to5.step

Part	Name	What to look for	Positions in the model	Buy	Each	Where to buy
P27	Handler magnet	12 V 25 mm lifting electromagnet (~2.5 kg hold) + MOSFET module + flyback diode	P27-1	1	£12	Amazon UK search shop
P28	Model beams	3D printed UB 305x165 at 1:5 (61 x 33 mm), 1 m long, with 0.5 mm steel strip glued on the top flange	P28-1	4	£4	print <i>make / print it</i>
P29	USB wide-angle camera (safety zone)	1080p UVC, 90-120 deg view (e.g. Logitech C920/C922 or an Arducam USB wide-angle)	P29-1	1	£50	Amazon UK, Pimoroni search shop
P30	CSI camera (close-up of the Cutter)	Arducam IMX219 for Jetson Orin Nano (22-pin cable), wide-angle lens	<i>not drawn (wiring, inside the control box, tools)</i>	1	£25	Arducam, RobotShop UK product page
P31	Powered USB 3 hub	4-port, own power supply	<i>not drawn (wiring, inside the control box, tools)</i>	1	£20	Amazon UK search shop
P32	Emergency stop station	yellow box, red 40 mm mushroom, turn to release, 2 NC contacts (e.g. Schneider Harmony XALK178 family)	P32-1, P32-2	1	£42	RS, Farnell, TME search shop
P33	Safety relay	Pilz PNOZ s3 24 V DC (dual channel, manual reset) or equivalent	<i>not drawn (wiring, inside the control box, tools)</i>	1	£180	RS, Farnell (new) / ~£75 used search shop
P34	Door interlock switch	non-contact coded magnetic safety switch, NC (e.g. Pilz PSEN or Schmersal BNS)	P34-1	1	£60	RS, Farnell search shop
P35	Reset push button	22 mm blue illuminated, 1 NO	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	RS, Amazon UK search shop
P36	Stack light	24 V LED tower, red / amber / green / blue	<i>not drawn (wiring, inside the control box, tools)</i>	1	£25	Amazon UK search shop
P37	4-channel relay module (opto-isolated)	3.3 V logic input, 24 V contacts	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	Amazon UK search shop
P38	IR break-beam sensor pair	5 mm IR break-beam (demonstrates the light-curtain input)	<i>not drawn (wiring, inside the control box, tools)</i>	1	£6	Pimoroni, The Pi Hut search shop
P39	Polycarbonate sheet 3 mm	front, door and ends (~1.5 m2) - polycarbonate, not acrylic (acrylic shatters)	P39-1	1.5 m2	£40	plastics stockist search shop
P40	Hinges + handle + 20x20 door frame	2 hinges, 1 handle, 2.4 m 20x20	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£25	Ooznest search shop
P41	24 V power supply	Mean Well LRS-150-24 (24 V 6.5 A) - motors, controller, safety circuit	<i>not drawn (wiring, inside the control box, tools)</i>	1	£16	RS (0424827) product page
P42	IEC mains inlet with switch and fuse	panel mount, 5 A fuse	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	RS, Farnell search shop
P43	DIN rail + terminals + fuse holders	35 mm DIN rail, 20 terminals, 4 x 5x20 fuse holders, end stops	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£30	RS, Farnell search shop
P44	Control box	ABS or steel enclosure ~300 x 200 x 150 with cable glands	P44-1	1	£30	RS, CPC search shop
P45	Wire + ferrules + crimp tool	0.75 and 1.5 mm2 flexible, ferrule kit	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£30	Amazon UK, CPC search shop
P46	Jetson Orin Nano Super (4 GB)	you already have it - docs/JETSON_SPECS.md	<i>not drawn (wiring, inside the control box, tools)</i>	1	£0	- <i>make / print it</i>
P47	PETG and PLA filament	1 kg each	<i>not drawn (wiring, inside the control box, tools)</i>	2	£20	Amazon UK, 3DJake search shop

Steel Beam Cutting Cell - 1:5 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 6 CAD: cad/prototype_1to5.step

Position list - what each position is and where it goes

x is along the machine (infeed end = 0), y across (front, the operator's side, is negative), z up from the table. Sizes in mm, centre of each part.

Position	Part	What it is	Where	x	y	z
P01-1	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - base rail 1500 mm	front (operator side), middle, 20 mm up	600	-300	20
P01-2	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - gantry rail 1500 mm (the bridges' wheels run on it)	front (operator side), middle, 660 mm up	600	-300	660
P01-3	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	front (operator side), infeed end, 340 mm up	-140	-300	340
P01-4	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	front (operator side), middle, 340 mm up	600	-300	340
P01-5	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	front (operator side), outfeed end, 340 mm up	1340	-300	340
P01-6	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - base rail 1500 mm	back, middle, 20 mm up	600	300	20
P01-7	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - gantry rail 1500 mm (the bridges' wheels run on it)	back, middle, 660 mm up	600	300	660
P01-8	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	back, infeed end, 340 mm up	-140	300	340
P01-9	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	back, middle, 340 mm up	600	300	340
P01-10	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - upright 600 mm	back, outfeed end, 340 mm up	1340	300	340
P01-11	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - cross member 580 mm	middle, infeed end, 20 mm up	-140	0	20
P01-12	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - cross member 580 mm	middle, infeed end, 660 mm up	-140	0	660
P01-13	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - cross member 580 mm	middle, outfeed end, 20 mm up	1340	0	20
P01-14	V-slot 20x40 aluminium extrusion	20x40 V-slot aluminium extrusion - cross member 580 mm	middle, outfeed end, 660 mm up	1340	0	660
P02-1	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed rail 1300 mm	front (operator side), middle, 150 mm up	600	-160	150
P02-2	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed rail 1300 mm	middle, middle, 150 mm up	600	-40	150
P06-1	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), infeed end, 170 mm up	100	-100	170
P07-1	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), infeed end, 158 mm up	100	-160	158
P07-2	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, infeed end, 158 mm up	100	-40	158
P06-2	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), middle, 170 mm up	300	-100	170
P07-3	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), middle, 158 mm up	300	-160	158
P07-4	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, middle, 158 mm up	300	-40	158
P06-3	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), middle, 170 mm up	500	-100	170
P07-5	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), middle, 158 mm up	500	-160	158
P07-6	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, middle, 158 mm up	500	-40	158
P06-4	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), middle, 170 mm up	700	-100	170
P07-7	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), middle, 158 mm up	700	-160	158
P07-8	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, middle, 158 mm up	700	-40	158
P06-5	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), middle, 170 mm up	900	-100	170
P07-9	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), middle, 158 mm up	900	-160	158
P07-10	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, middle, 158 mm up	900	-40	158
P06-6	Bed rollers	Bed roller: 20 mm aluminium tube on an 8 mm shaft, 2 x 608 bearings	front (operator side), outfeed end, 170 mm up	1100	-100	170
P07-11	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	front (operator side), outfeed end, 158 mm up	1100	-160	158
P07-12	Roller brackets	3D-printed roller bracket (cad/prototype: print in PETG)	middle, outfeed end, 158 mm up	1100	-40	158
P02-3	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	front (operator side), infeed end, 90 mm up	-40	-160	90
P02-4	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	middle, infeed end, 90 mm up	-40	-40	90
P02-5	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	front (operator side), middle, 90 mm up	600	-160	90
P02-6	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	middle, middle, 90 mm up	600	-40	90
P02-7	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	front (operator side), outfeed end, 90 mm up	1240	-160	90
P02-8	V-slot 20x20 aluminium extrusion	20x20 V-slot - bed leg	middle, outfeed end, 90 mm up	1240	-40	90
P09-1	Scrap tray	Scrap tray: 3D-printed or folded aluminium sheet	front (operator side), middle, 45 mm up	600	-100	45
P08-1	Outfeed table board	Outfeed table: 6 mm plywood / MDF strip on 20x20 legs	back, middle, 177 mm up	600	170	177
P28-1	Model beams	Model beam: UB 305x165x40 at 1:5, 3D printed (see docs/SCALE_MODEL.md)	front (operator side), middle, 210 mm up	520	-100	210
P03-1	V-slot 20x60 aluminium extrusion	Handler bridge: 20x60 V-slot 660 mm	middle, infeed end, 710 mm up	150	0	710
P10-1	V-wheel gantry plate kits	Handler bridge end: V-slot gantry plate with 4 V-wheels	front (operator side), infeed end, 698 mm up	150	-300	698
P10-2	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	front (operator side), infeed end, 685 mm up	120	-300	685

Steel Beam Cutting Cell - 1:5 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 7 CAD: cad/prototype_1to5.step

Position	Part	What it is	Where	x	y	z
P10-3	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	front (operator side), infeed end, 685 mm up	180	-300	685
P10-4	V-wheel gantry plate kits	Handler bridge end: V-slot gantry plate with 4 V-wheels	back, infeed end, 698 mm up	150	300	698
P10-5	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	back, infeed end, 685 mm up	120	300	685
P10-6	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	back, infeed end, 685 mm up	180	300	685
P11-1	NEMA 17 stepper motor	Handler X motor: NEMA 17 (42x42x48) driving both sides through an 8 mm cross shaft and GT2 belts	front (operator side), infeed end, 741 mm up	191	-236	741
P17-1	8 mm cross shafts + pillow blocks	8 mm cross shaft with GT2 20T pulleys at both ends	middle, infeed end, 741 mm up	191	0	741
P10-7	V-wheel gantry plate kits	Handler carriage: V-slot gantry plate on the bridge (Y axis)	middle, infeed end, 688 mm up	150	0	688
P11-2	NEMA 17 stepper motor	Handler Y motor: NEMA 17 + GT2 belt along the bridge	middle, infeed end, 661 mm up	186	0	661
P02-9	V-slot 20x20 aluminium extrusion	Handler Z axis: 20x20 V-slot 300 mm on a T8 lead screw (slides through the carriage)	middle, infeed end, 646 mm up	150	0	646
P12-1	NEMA 17 with integrated T8 lead screw	Handler Z motor: NEMA 17 with integrated T8 x 2 mm lead screw	middle, infeed end, 714 mm up	91	0	714
P13-1	Z motor bracket (3D printed)	Handler Z motor bracket (3D printed)	middle, infeed end, 688 mm up	102	0	688
P12-2	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead screw, 300 mm	middle, infeed end, 593 mm up	91	0	593
P14-1	Lead-screw nut block (3D printed)	Handler lead-screw nut block on the Z axis (3D printed + brass T8 nut)	middle, infeed end, 503 mm up	110	0	503
P23-1	Arm mount plate (3D printed)	Handler arm mount plate (3D printed)	middle, infeed end, 493 mm up	150	0	493
P22-1	SO-101 6-servo arm kit (follower)	Handler arm: SO-101 type 6-servo arm kit (STS3215 servos) - base	middle, infeed end, 465 mm up	150	0	465
P22-2	SO-101 6-servo arm kit (follower)	Handler arm: upper arm (~110 mm)	middle, infeed end, 385 mm up	150	0	385
P22-3	SO-101 6-servo arm kit (follower)	Handler arm: forearm (~135 mm)	middle, infeed end, 318 mm up	206	0	318
P22-4	SO-101 6-servo arm kit (follower)	Handler arm: wrist	middle, middle, 286 mm up	275	0	286
P27-1	Handler magnet	Handler magnet: 12 V 25 mm lifting electromagnet (holds 2.5 kg)	middle, middle, 253 mm up	275	0	253
P03-2	V-slot 20x60 aluminium extrusion	Cutter bridge: 20x60 V-slot 660 mm	middle, middle, 710 mm up	900	0	710
P10-8	V-wheel gantry plate kits	Cutter bridge end: V-slot gantry plate with 4 V-wheels	front (operator side), middle, 698 mm up	900	-300	698
P10-9	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	front (operator side), middle, 685 mm up	870	-300	685
P10-10	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	front (operator side), middle, 685 mm up	930	-300	685
P10-11	V-wheel gantry plate kits	Cutter bridge end: V-slot gantry plate with 4 V-wheels	back, middle, 698 mm up	900	300	698
P10-12	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	back, middle, 685 mm up	870	300	685
P10-13	V-wheel gantry plate kits	V-wheel (Delrin, 625 bearings)	back, middle, 685 mm up	930	300	685
P11-3	NEMA 17 stepper motor	Cutter X motor: NEMA 17 (42x42x48) driving both sides through an 8 mm cross shaft and GT2 belts	front (operator side), middle, 741 mm up	941	-236	741
P17-2	8 mm cross shafts + pillow blocks	8 mm cross shaft with GT2 20T pulleys at both ends	middle, middle, 741 mm up	941	0	741
P10-14	V-wheel gantry plate kits	Cutter carriage: V-slot gantry plate on the bridge (Y axis)	middle, middle, 688 mm up	900	0	688
P11-4	NEMA 17 stepper motor	Cutter Y motor: NEMA 17 + GT2 belt along the bridge	middle, middle, 661 mm up	936	0	661
P02-10	V-slot 20x20 aluminium extrusion	Cutter Z axis: 20x20 V-slot 300 mm on a T8 lead screw (slides through the carriage)	middle, middle, 701 mm up	900	0	701
P12-3	NEMA 17 with integrated T8 lead screw	Cutter Z motor: NEMA 17 with integrated T8 x 2 mm lead screw	middle, middle, 714 mm up	841	0	714
P13-2	Z motor bracket (3D printed)	Cutter Z motor bracket (3D printed)	middle, middle, 688 mm up	852	0	688
P12-4	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead screw, 300 mm	middle, middle, 620 mm up	841	0	620
P14-2	Lead-screw nut block (3D printed)	Cutter lead-screw nut block on the Z axis (3D printed + brass T8 nut)	middle, middle, 558 mm up	860	0	558
P23-2	Arm mount plate (3D printed)	Cutter arm mount plate (3D printed)	middle, middle, 548 mm up	900	0	548
P22-5	SO-101 6-servo arm kit (follower)	Cutter arm: SO-101 type 6-servo arm kit (STS3215 servos) - base	middle, middle, 520 mm up	900	0	520
P22-6	SO-101 6-servo arm kit (follower)	Cutter arm: upper arm (~110 mm)	middle, middle, 440 mm up	900	0	440
P22-7	SO-101 6-servo arm kit (follower)	Cutter arm: forearm (~135 mm)	middle, outfeed end, 373 mm up	956	0	373
P22-8	SO-101 6-servo arm kit (follower)	Cutter arm: wrist	middle, outfeed end, 341 mm up	1025	0	341
P26-1	Cutter 'torch' for the prototype	Cutter 'torch' for the prototype: pen or 5 mW laser pointer in a 3D-printed holder (marks the cut lines)	middle, outfeed end, 286 mm up	1025	0	286
P44-1	Control box	Control box: Jetson Orin Nano, FluidNC 6-axis board, 24 V and 12 V supplies, fuses, safety relay	middle, outfeed end, 80 mm up	1480	0	80
P32-1	Emergency stop station	Emergency stop: 22 mm red mushroom, 2 NC contacts, yellow box (BS EN ISO 13850)	front (operator side), outfeed end, 35 mm up	1445	-345	35
P32-2	Emergency stop station	Emergency stop head	front (operator side), outfeed end, 82 mm up	1445	-345	82
P39-1	Polycarbonate sheet 3 mm	Enclosure door: 3 mm polycarbonate on 20x20 frame, with a safety interlock switch	front (operator side), middle, 340 mm up	400	-350	340
P34-1	Door interlock switch	Door interlock switch (magnetic coded, NC) - opens = protective stop	front (operator side), middle, 320 mm up	740	-361	320
P29-1	USB wide-angle camera (safety zone)	Camera: USB wide-angle (or IMX219 CSI) on a 20x20 post - YOLO watches the cell	front (operator side), infeed end, 760 mm up	-190	-300	760
P02-11	V-slot 20x20 aluminium extrusion	20x20 V-slot camera post	front (operator side), infeed end, 380 mm up	-190	-300	380

Steel Beam Cutting Cell - 1:5 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 04 Oct 2026 Units: mm Scale: not to scale	Sheet 8 CAD: cad/prototype_1to5.step